



Avoiding Algorithmic Bias

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Student Materials

- Long version of worksheet with questions and EU Guidelines
 - web
 - PDF
- Scenarios only version of worksheet
 - web
 - PDF
- EU AI Guidelines slide deck
 - web

Overview

Ethics background required: Ideally, students should be familiar with the 4 ethical frameworks presented in the first year curriculum (virtue ethics, deontology, consequentialism, analogies). If they (or the instructor) are not, a brief summary is provided in the reflection.

Subject matter referred to in this lab: Algorithms

Placement in overall ethics curriculum:

- Academic year: Year 2-4 core course that talks about algorithms or machine learning.
- Recommended previous labs: First year ethics curriculum
- Recommended follow-up labs: No direct follow-up labs

Time required:

- In class: 30 minutes (online version is also included)

Learning objectives:

- Students are introduced to or reminded of the dangers of algorithmic bias in predictive algorithms
- Students are introduced to a deontological set of guidelines to evaluate an algorithm for bias



- Students apply the guidelines to actual algorithms and practice this evaluation
- Students review the four ethical frameworks and evaluate the benefit of using each in regard to evaluating predictive algorithms

Ethical issues to be considered: Algorithmic bias, diversity, transparency

Flow

- Students read and talk about algorithms with issues
- Students are introduced to the EU Ethics Guidelines for Trustworthy AI [1]
- Students evaluate the algorithms previously presented based on these guidelines

Preparation

- Read through the entire lab.
- Print enough of each of the 3 scenarios to distribute one each to group of 2-4 students. Or give all three scenarios to each group and assign one of the scenarios to each group.
- Prepare a PowerPoint (or similar) slide to project with the 7 EU Guidelines on it.

Guide for Instructors

Lesson plan

Introduction (5 minutes)

Ask students for the definition of “algorithm.” (Something like a step-by-step approach to use as a guideline for producing a program that the computer can use to solve a problem)

Remind students that their college application was probably subject to some sort of algorithm to determine whether they were admitted. Ask class to suggest what some of the steps might have been in the decision making process. (This is similar to question 1 from the worksheet.)

Here are some possibilities:

- Is the GPA > X (X depends on institution)
- Is the SAT/ACT > X
- 5 points for each area of community service

To students: Generally, these types of algorithms are point-based. Why would a point-based method be adopted? What are the benefits? (probably something about objectivity)

Do you see any problems with the point-based method? (possible responses: not everything can be boiled down to numbers, how to decide how many points something is worth,



Activity 1 (5 min)

Facilitate the division of students into groups of 2-4 members. Each group should be given one of the scenarios in the worksheet. Direct the students to read the scenario as a group and discuss what went right and what went wrong.

- Scenario 1: Market crash of 2008 (algorithms as objective agents)
- Scenario 2: Automating Hiring (providing feedback to an algorithm)
- Scenario 3: Modeling Churn (seeking fairness)

Preparation for activity 2 (5 min)

Instructor should read or summarize after the previous group activity is finished:

Point-based algorithms for making decisions are a rudimentary form of Artificial Intelligence. Kaplan and Haenlein recently defined AI as “a system’s ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation.” Our example of a point-based algorithm for college admission is used to predict who will most likely succeed at a particular college or university. Similarly, each of the scenarios that you read involved simple predictive algorithms.

We hear frequently of the challenges with AI and even simple predictive algorithms share the same issues. Recently, the EU came up with a set of guidelines to help diminish the chance of ethical issues in such algorithms. As we go through these guidelines, think back on our college admissions example and the example that you read in your group and see if any of them would be helpful to consider if you were in charge of developing such an algorithm. Be prepared to discuss your observations with your group.

Note to instructor: The EU Ethics Guidelines for Trustworthy AI should be made available to students so they can follow along.

- One easy approach is to project the guidelines on a screen for the class. You can use this web-based slide deck.
- It is a good idea for students to participate in the reading (assign 1 guideline to a student to read) so that they are more engaged.
- These also appear on the student worksheet.

Note to instructor: Definition of AI. There are some weaknesses to the definition of AI given by Kaplan and Haenlein. In particular, many of the terms used in the definition do not have



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self-evident meanings, so it would probably be difficult to use this definition to decide whether something is or is not an example of AI. Try not to focus on the issue of defining AI – keep the focus on the scenarios and the ethical issues they raise.

Activity 2 (5 min)

Students discuss with their group observations that they made regarding the potential benefits of applying 1 or more of the guidelines to the college admissions scenario or to the scenario read as a group.

Reflection (10 min)

Have each group report on one or more of the guidelines that would have helped in the design of either the college admission scenario or the scenario that they read. If they choose to use their group scenario, have the group first summarize the scenario. It is probably best to ask if other groups have comments on that same scenario before moving on.

Ask the students: Who can tell me which ethical framework is being used by applying a set of guidelines to vet a predictive algorithm? (virtue ethics, deontological, analogies, consequentialism?) If they need a refresher:

- Virtue ethics: making a decision based on whether the outcome upholds a set of virtues (honesty, compassion, patience, hard-work are some examples)
- Deontological ethics: making decisions based on a set of rules (this is the expected choice)
- Analogies: making a decision based on a similar situation where the ethicality of the similar situation was widely accepted
- Consequentialism: making a decision based on what is best for the majority

Ask the students if they believe that this was an appropriate framework to use in this case. Why or why not?

Possible answers:

- Probably good if used only as guidelines. It's a good starting place
- Virtue ethics might be a good secondary approach, and some of the virtues (respect for diversity, for example) are included in the guidelines
- Analogies probably would not be a good approach as it is not a single decision that must be made and it would be very difficult to arrive at an appropriate analogy
- Consequentialism is probably what caused the problem in the first place – designers would figure that it would work for most people.

Assessment

- Assess the deontological approach to evaluating an algorithm for ethicality compared to the virtue ethics, analogy, or consequentialist approach. Is it better, worse, explain.



- Provide 2 of the guidelines listed in the EU Ethics Guidelines for Trustworthy AI that we talked about in class

- Summarize one of the examples of predictive algorithms that exhibited bias that were presented in class. Specifically state the bias, and explain how it could have been avoided.

Sources

All of the scenarios were excerpted from C. O'Neil [2]. Cathy O'Neil is also the author of Weapons of Math Destruction.

Principles are directly from the text of L. Floridi [1], compiled by the High-Level Expert Group on Artificial Intelligence set up by the European Commission.

AI definition is from M. Kaplan Andreas; Haenlein [3]

Additional reading

N. T. Lee, P. Resnick, and G. Barton [4]

References

Bibliography

- [1] L. Floridi, "Establishing the rules for building trustworthy AI," Nature Machine Intelligence, vol. 1, no. 6, pp. 261–262, 2019, [Online]. Available: https://ai.bsa.org/wp-content/uploads/2019/09/AIHLEG/_EthicsGuidelinesforTrustworthyAI-ENpdf.pdf
- [2] C. O'Neil, "How Algorithms Rule Our Working Lives." [Online]. Available: <https://www.theguardian.com/science/2016/sep/01/how-algorithms-rule-our-working-lives>
- [3] M. Kaplan Andreas; Haenlein, "Siri, Siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence," Business Horizons, vol. 62, no. 1, pp. 15–25, Jan. 2019.
- [4] N. T. Lee, P. Resnick, and G. Barton, "Algorithmic bias detection and mitigation: Best practices and policies to reduce consumer harms." [Online]. Available: <https://www.brookings.edu/research/algorithmic-bias-detection-and-mitigation-best-practices-and-policies-to-reduce-consumer-harms/>